

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Name

ShadowEvent AI

Problem Statement

Traffic authorities primarily respond to incidents after congestion has already formed.

Historical traffic incident records contain recurring spatial and temporal patterns that can be used to proactively identify high-risk periods and locations.

The objective of ShadowEvent AI is to discover recurring traffic disruptions ("Shadow Events"), generate proactive risk intelligence, and provide operational decision support using historical incident data.

Project Goal

Build a web-based intelligence platform that:

1. Discovers recurring traffic incident patterns.
2. Generates a weekly risk calendar.
3. Finds similar historical incidents.
4. Visualizes high-risk zones on a map.
5. Creates a searchable traffic knowledge repository.

The system must operate primarily on the ASTRAM historical incident dataset.

Scope

Included

- Historical incident analysis
- Pattern mining
- Recurring hotspot discovery
- Risk scoring
- Event similarity search
- Interactive mapping

- Dashboard visualization

Excluded

- Real-time traffic prediction
- Digital twin simulation
- Reinforcement learning
- Traffic flow simulation
- Signal optimization
- Dynamic routing

These remain future extensions.

User Personas

Traffic Control Officer

Needs:

- High-risk locations
 - Risk forecasts
 - Historical incident context
-

Traffic Operations Manager

Needs:

- Weekly planning information
 - Hotspot trends
 - Resource prioritization
-

City Administrator

Needs:

- High-level dashboards
 - Recurring congestion insights
 - Data-driven planning reports
-

Core Modules

MODULE 1

Shadow Event Discovery Engine

Objective

Identify recurring incident patterns.

Inputs

Historical incident records

Fields:

- Timestamp
 - Latitude
 - Longitude
 - Incident Type
 - Location
-

Processing

Extract:

- Day of week
- Hour of day
- Month

Aggregate:

Location × Day × Hour

Compute:

- Incident count
 - Recurrence frequency
 - Probability score
-

Output

Shadow Event Record

Example:

```
{ location: "Silk Board", day: "Thursday", hour_range: "18-21", incident_probability: 0.73 }
```

Success Criteria

System automatically discovers recurring hotspots.

MODULE 2

Risk Calendar Generator

Objective

Generate proactive risk forecasts.

Inputs

Output from Shadow Event Discovery Engine

Processing

Rank all discovered Shadow Events by risk score.

Create:

- Daily view
 - Weekly view
-

Output

Weekly Risk Calendar

Example:

Monday

Airport Road Risk: High

Thursday

Silk Board Risk: High

Success Criteria

Users can identify future high-risk periods.

MODULE 3

Event Similarity Engine

Objective

Retrieve historical incidents similar to a selected event.

Inputs

Selected event

Features:

- Location
 - Incident Type
 - Day
 - Time
-

Processing

Feature Encoding

Nearest Neighbor Search

KNN Similarity

Output

List of similar incidents

Example:

Current Event



Top 10 Similar Historical Events



Observed Outcomes

Success Criteria

Relevant historical incidents returned.

MODULE 4

Impact Zone Visualization

Objective

Visualize risk areas on a map.

Inputs

Incident locations

Shadow Events

Hotspots

Processing

Generate:

- Point markers
 - Cluster layers
 - Heatmaps
-

Output

Interactive map

Displays:

- Event locations
 - Hotspots
 - Risk zones
-

Success Criteria

User can visually inspect risk concentrations.

MODULE 5

Traffic Knowledge Repository

Objective

Provide searchable historical intelligence.

Inputs

All processed incident records

Features

Search by:

- Location
- Incident Type
- Day
- Time

Filter by:

- Severity
 - Frequency
 - Date range
-

Output

Historical incident explorer

Success Criteria

Users can retrieve relevant historical information quickly.

Data Pipeline

Raw ASTRAM CSV

↓

Data Cleaning

↓

Feature Engineering

↓

Pattern Mining

↓

Risk Scoring



Database Storage



API Layer



Frontend Dashboard

Suggested Database Schema

Table: incidents

Columns:

id

timestamp

latitude

longitude

location

incident_type

day_of_week

hour

month

Table: shadow_events

Columns:

id

location

day_of_week

hour_bucket

incident_count

probability_score

risk_level

Table: similarity_index

Columns:

id

incident_id

embedding_vector

Backend APIs

GET /shadow-events

Returns all discovered Shadow Events

GET /risk-calendar

Returns weekly risk calendar

GET /hotspots

Returns hotspot locations

GET /similar-events/{id}

Returns similar historical incidents

GET /map-data

Returns heatmap and marker data

GET /search

Search historical records

Frontend Pages

1. Dashboard

Metrics:

- Total Incidents
 - Total Shadow Events
 - Top Hotspots
-

1. Risk Calendar

Weekly view

Daily view

Risk rankings

1. Shadow Event Explorer

Location search

Pattern visualization

1. Similarity Explorer

Select incident

View similar events

1. Interactive Map

Heatmap

Markers

Clusters

Filters

1. Knowledge Repository

Search

Filters

Historical analytics

Recommended Tech Stack

Frontend

React

TypeScript

Tailwind

Leaflet

Backend

FastAPI

Python

ML / Analytics

Pandas

NumPy

Scikit-Learn

Database

PostgreSQL

Maps

OpenStreetMap

Leaflet

Hackathon MVP Priority

Priority 1 (Must Build)

✓ Shadow Event Discovery

✓ Risk Calendar

✓ Hotspot Detection

Priority 2

✓ Interactive Map

✓ Knowledge Repository

Priority 3

✓ Similarity Search

Deliverables

1. Working Dashboard

2. Weekly Risk Calendar
 3. Shadow Event Discovery Module
 4. Hotspot Visualization Map
 5. Similarity Search Module
 6. Searchable Incident Repository
 7. Final Presentation
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Agent Execution Goal

Given the ASTRAM dataset, automatically:

1. Analyze schema.
2. Clean and preprocess data.
3. Build Shadow Event discovery pipeline.
4. Generate hotspot rankings.
5. Generate risk calendar.
6. Build similarity search index.
7. Expose APIs.
8. Create dashboard UI.
9. Create map visualizations.
10. Prepare demo-ready outputs.